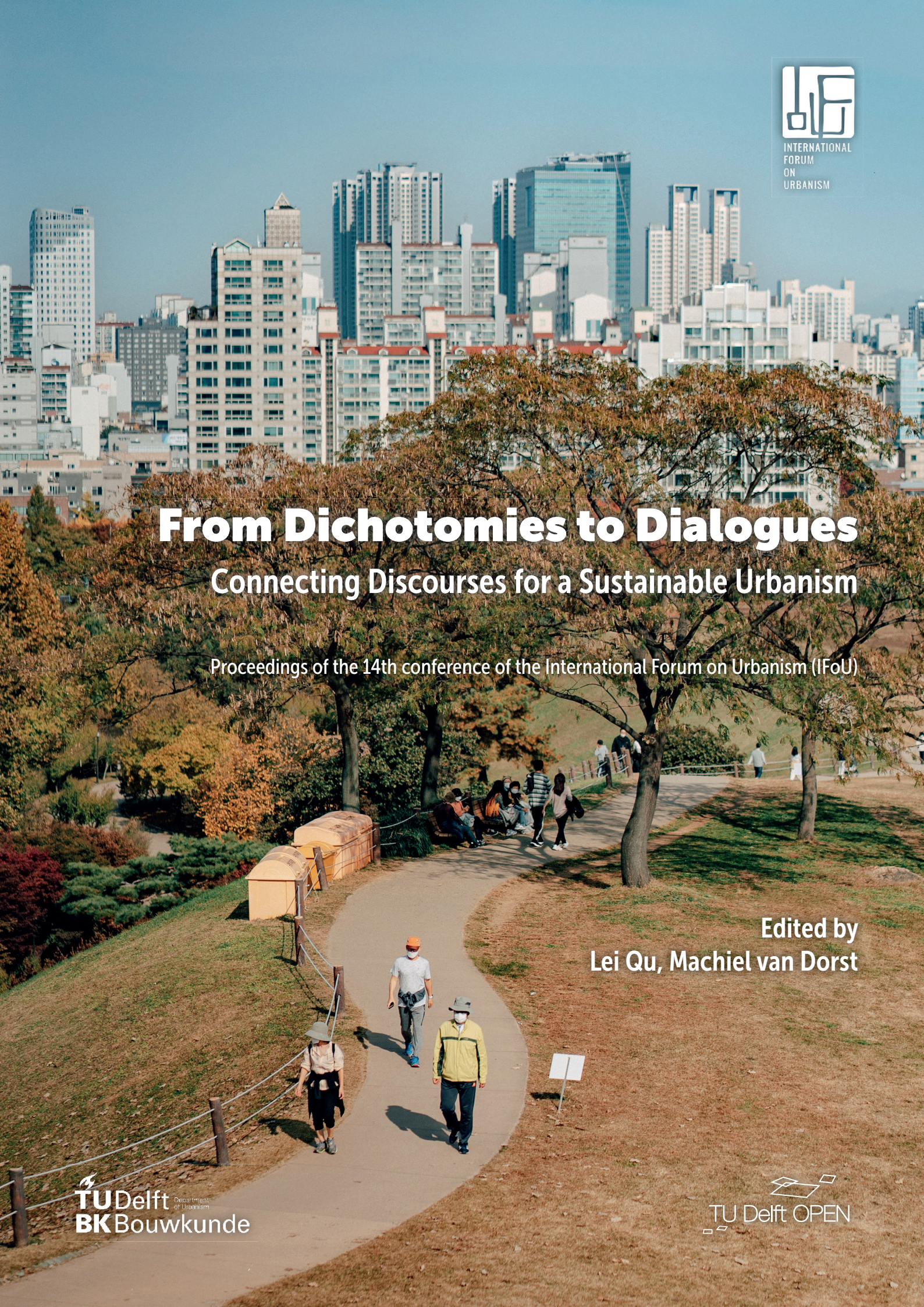


From Dichotomies to Dialogues

Connecting Discourses for a Sustainable Urbanism

Proceedings of the 14th conference of the International Forum on Urbanism (IFoU)

Edited by
Lei Qu, Machiel van Dorst

TU Delft Department of Urbanism
BK Bouwkunde

TU Delft OPEN

Type of the paper: Peer-reviewed Conference Paper / Short Paper

Diversity of Built Form: Is It and Its Relations with Other Dimensions of Diversity Well-Measured?

Zhuoshu He [1]*, Ye Zhang [2]

[1] National University of Singapore; zhuoshuhe@u.nus.edu; https://orcid.org/0000-0003-0499-7111
[2] National University of Singapore; akizy@nus.edu.sg
* Corresponding author

Research highlights

1) Conduct a literature review to evaluate the current status in quantifying diversity of urban form.

2) Summarize multidisciplinary definition and measurements of diversity, setting a triple-concept framework comprising variety, balance and disparity for evaluating existing studies.

3) Layout existing studies about measuring built-form diversity and assessing its relations with other socio-spatial factors.

4) Results show that no study measures built-form diversity considering the comprehensive diversity definition, and overall weak and inconsistent associations between built-form diversity and other socio-spatial factors.

Keywords: urban diversity; urban form; urban morphology; measurement; literature review

Names of the track editors:
Birgit Hausleitner
Leo van den Burg
Akkelies van Nes

Names of the reviewers:
Birgit Hausleitner
Jeroen van Ameijde

Journal: The Evolving Scholar

DOI:10.24404/6156d4c081455e00084986e3

Submitted: 01 October 2021
Accepted: 01 June 2022
Published: 30 August 2022

Citation: He, Z. & Zhang, Y. (2021). Diversity of Built Form: Is It and Its Relations with Other Dimensions of Diversity Well-Measured? The Evolving Scholar | IFoU 14th Edition.

This work is licensed under a Creative Commons Attribution BY (CC BY) license.
©2021 [He, Z. and Zhang, Y.] published by TU Delft OPEN on behalf of the authors.

1. Introduction

Diversity is one of the most essential values in contemporary urban planning. It is able to enhance place vitality and social equity, especially for many regions suffered from separation for decades (Talen & Lee, 2018). Urban diversity is generally regarded as having physical and social dimensions. The former is various physical design, mixes of uses, and the expanded public realm, while the latter consists of mixes of multiple class, ethnic, and cultural groups exercising their rights (Bobkova, Marcus, & Berghauser Pont, 2017; Fainstein, 2005; Jacobs, 1961; Talen, 2006b). Physical diversity does not directly generate social diversity, but it does provide spatial conditions for social diversity to thrive and sustain (Talen, 2006b). Some researchers even believe that given the fluid nature of social diversity, providing diverse forms in the built environment may be more successful than policies directly aim at integration (Stanley, 2012).

Measuring the diversity of built form is crucial for resolving the above debates about the effects of physical diversity on the overall urban diversity. In addition, it is important to distinguish which spatial variables are important for socioeconomic diversity and in what combinations (Bobkova et al., 2017). The prerequisite is properly measuring physical diversity properly. Secondly, a diverse built environment is a common goal in urban design. The plot, street, block, etc. are fundamental components that urban designers work on. Currently, existing urban design protocols and guidelines (e.g., ASBEC, 2017) are too general in providing guidance for these components. Better measurement and assessment will provide stronger references to inform urban design guidelines. Thirdly, urban forms can be studied independently by studying the mathematical relationships between the characteristics of form (Vialard, 2013). It helps enhance knowledge on urban spatial structure and develop methods that adapt urban analytics better.

Scholars pointed out that methods and techniques for measuring diversity of built form are not sophisticated and need to be investigated in more detail (Marcus, 2010; Talen, 2006c; Talen & Lee, 2018). Compared to built-form measurements on dimension, intensity, and connectivity, diversity is underrepresented (Fleischmann, Romice, & Porta, 2020). Additionally, existing diversity measures originate from multiple subjects (e.g., ecology, economics, and psychology). Whether and how they could be applicable to urban analytics require more exploration (Salat, Bourdic, & Nowacki, 2010).

Nonetheless, what is the current status on research about measuring built-form diversity and assessing its relations with other socio-spatial factors? Comprehensive review on this topic is still rare, which obstructs further discussion and exploration on this topic. Therefore, we conduct a literature review in order to: 1) summarise a multidisciplinary definition and measures of diversity to set a framework for filtering existing quantitative research about built-form diversity; and 2) layout measurements of built-form diversity, and assessments of relations between built-form diversity and other socio-spatial factors.

2. Methods of Review

Our review starts with technical definitions and measures on diversity. Here we do not confine the scope in urban morphology or urban studies, but include multidisciplinary literature. It is for setting a framework for filtering relevant literature.

Because the terminology in research about both diversity and urban form are inconsistent and loose, directly identifying literature by keywords seems difficult. We apply an iterative snowball approach combining backward and forward traces of reference. One starting point is a most recent review about built-form measurements (Fleischmann et al., 2020), which summarises four quantitative measurements on diversity of urban form (Bobkova et al., 2017; Feliciotti, 2018; Louf & Barthelemy, 2014; Song, Gordon-Larsen, & Popkin, 2013). Another starting point is literature citing a classic work about urban diversity by Talen (2008). Whenever we find literature related to measuring built-form diversity, it is included in the database, and its reference is screened. Correspond to Fleischmann et al. (2020), for studies about non-morphological elements (housing, land use, and transportation), we note them but do not include them in this review.

Then we extract measurements on diversity of built form, which include the following contents. 1) Element: the spatial unit (or grain) whose features are measured, including building, plot, street and block. 2) Feature: a characteristic of the element that is measured (e.g., area). 3) Extent: the aggregated area in which a diversity measurement is computed and the spatial unit that diversity value is assigned (e.g., 500 m network accessible district of Census Tracts). 4) Measure: the formula or desiderata to compute diversity (e.g., Simpson index).

3. Results

3.1. Multidisciplinary definitions and measures of diversity

The definitions of diversity are manifold and ambiguous (Bobkova et al., 2017; Talen, 2006a). Many urban studies do not specify a definition of diversity, but take it as a notion instead. Some researchers define it through how it is intrinsically produced, like Page (2010)'s diversity "of types," "within a type," and "of composition," and Salat et al. (2010)'s diversity in "similar objects," "spatial distribution," and "objects of different scales."

We highlight diversity definitions that describe the performance and lead to measures. It is assumed that a system encompasses plenty of objects, and can be partitioned into categories according to attributes. The most popular definition relates to ecology and combines two concepts, richness and evenness (Magurran, 2005; McGarigal, Cushman, Neel, & Ene, 2002). Richness is the number of categories, while evenness is their relative abundance. However, Junge (1994) and Stirling (2007) advocate a more comprehensive definition, comprising variety, balance, and disparity. Variety is the synonym of richness. Balance is the apportionment across categories, capturing both

evenness in ecology and concentration in economy. Variety and balance are characterised by categories. Disparity refers to the manner and degree in which objects are distinguished. It is on this basis that a taxonomy of objects is defined and partitioned, hence, it could also represent inter-categorial heterogeneity.

Related measures can be placed in the above three concepts. Variety can be simply measured by categorial count. It can be further expressed as joint or non-overlapping types (Page, 2010). Measures of balance are mainly Shannon Entropy and Simpson Index. There are some variants, including the Herfindahl–Hirschman Index, Gini Index, etc. The weakness is that they treat all categories as equal (Page, 2010). Usually, even distribution is not the reality nor the optimum (Salat et al., 2010). Some researchers instead compare categorial percentages between sub-district and entire areas, which imply some kinds of weighting, like the Dissimilarity Index and Neighborhood Diversity Index (Maly, 2000; Song, Merlin, & Rodriguez, 2013). For disparity, the simplest measures are variation measures, like standard deviation (SD) and coefficient of variation (CV) (Junge, 1991). Balance measures fail to take into account the extent of differences between categories, which could be addressed by distance measures. Weitzman (1992) used a set of pairwise distances of objects to perform desiderata of taxonomy. However, distance measures rely on the sample – when more objects are added, the distances are likely to change (Page, 2010). For some attributes, distance may not exist between objects (Page, 2010). Additionally, Junge (1991; 1994) used projection models to measure similarity/dissimilarity (homogeneity/heterogeneity) of objects and apportionment between categories.

3.2. Existing studies measuring diversity of built form

Based on the triple-concept framework, we select and summarise built-form measurements that measure variety, balance, or disparity. A few studies claim that they intend to quantify built-form diversity, and often take SD, CV, Simpson index, or Shannon entropy as proxies for diversity. More studies rarely mention diversity, but measure variety, balance, or disparity of built form to some extent, so they are also included (Table 1). Throughout both kinds of studies, no one covers variety, balance, and disparity at the same time, let alone integrating them.

Within studies measuring variety and balance, Bobkova et al. (2017) and Martino, Girling, and Lu (2021) emphasize diversity the most. For Bobkova et al. (2017), diversity is a starting point and is linked to the plot system based on the Spatial Capacity, so the aim is shifted to morphological measures for the plot. The only measurement that directly measures diversity is Simpson Index of plot area within a 500 m network accessible district (the value is assigned to the plot). They infer that diverse plot area is important for a fertile ground with high plot accessibility when it scales up from a local center to a more globally performing center. They replace this measurement with number of accessible plots in the subsequent research (Bobkova, Berghauser Pont, & Marcus, 2019). Similarly, in Martino et al. (2021), variables measuring diversity of built form are Simpson index of building age, plot area, and dwelling type.

Those measuring disparity often employ clustering and multivariate analysis for the following aims. Some studies aim to classify built-form elements into typologies. For example, Bobkova et al. (2019) explore plot typologies based on three attributes. Some aim to classify cities or districts through clustering. For instance, Boeing (2019) classifies 100 cities based on eight attributes on street configuration. Here, cities are the elements and their pairwise distances or disparity are measured. Some other studies aim to simplify built-form criteria applying multivariate analysis, where clustering of studied districts plays as validation of the reduced set of variables (Dibble et al., 2017; Song, Gordon-Larsen, et al., 2013). It is claimed to be measuring diversity by Salazar-Llano (2019). However, in all these situations, the balance of elements and features within certain extents (e.g., neighborhoods) is missing to measure diversity comprehensively.

Besides, most frequent elements and features are building age and plot area, as what Jacobs (1961) advocates. There are two ways to catch the extent – using exclusive

boundary (e.g., census district, sanctuary area and grids), or taking into account surrounding area (e.g., buffer, accessible network district). Most studies set the extent on the neighborhood or equivalent scale, corresponding to the argument that district (neighborhood) scale enables examination of diversity and social mixing (Bourdic, Salat, & Nowacki, 2012; Talen & Lee, 2018). Only Martino et al. (2021) compare different scales using five-radius network accessible districts of the census dissemination area.

Table 1. Measurements on diversity of built form

Element	Measurements of Variety, Balance	Measurements of Disparity
Building	• Simpson Index of building age within 400 m, 800 m, 1600 m, 3200 m and 4800 m network accessible districts of the census district (Martino et al., 2021) • Shannon Entropy of building floor area within all plots abutting street segment (Ewing & Clemente, 2013) • Shannon Entropy of building height within town (Salazar-Llano, 2019)	• SD (Z score) of building age within 200m*200m grid (Powe, Mabry, Talen, & Mahmoudi, 2016; Preservation Green Lab, 2014), 100m*100m grid (Delclòs-Alió & Miralles-Guasch, 2018) or census district (Rosner & Curtin, 2015; Sung, Lee, & Cheon, 2015) • K-means cluster of buildings within 5 cities based on floor area ratio and building density in 500 m network accessible district (Berghauser Pont, Stavroulaki, Bobkova, et al., 2019)
Plot	• Simpson index of plot area within 500m network accessible district (Bobkova et al., 2017; Feliciotti, 2018) or 400 m, 800 m, 1600 m, 3200 m, 4800 m network accessible districts of census district (Martino et al., 2021)	• K-means cluster of plots within five cities based on average plot area, average ratio of plot area, and minimum bounding rectangle area and average ratio of plot frontage and perimeter in 500 m network accessible district (Bobkova et al., 2019) • SD of land patch area within 1 km, 3 km, 5 km, 8 km buffer (Song, Gordon-Larsen, et al., 2013)
Block	-	• CV of block area, ratio of same-size-circle diameter and circumscribed-circle diameter, and ratio of block perimeter and same-size-square perimeter within adjacent blocks (Vialard, 2013) • Hierarchical cluster of 131 cities based on block area and ratio of block area and circumscribed circle area (Louf & Barthelemy, 2014)
Street	• Shannon entropy of street length and street orientation within town (Salazar-Llano, 2019) or city (Boeing, 2019)	• K-means cluster of streets within five cities based on angular betweenness (Berghauser Pont, Stavroulaki, Bobkova, et al., 2019) • Hierarchical cluster of 100 cities based on street orientation, length, circuity, and proportion of different nodes (Boeing, 2019)
Mixed elements	-	• Hierarchical cluster of 45 sanctuary areas based on 207 attributes and a subset of nine attributes on building, plot, and block, which are derived from multivariate analysis (Dibble et al., 2017) • Hierarchical cluster of 20,467 neighborhoods based on 27 attributes on land patch and street and subsets of attributes that are highest loading on factors within 1 km, 3 km, 5 km, 8 km buffer (Song, Gordon-Larsen, et al., 2013)

3.3. Quantitative evidence relating built-form diversity with other factors

Within quantitative studies about built-form diversity, evaluation of its relations with functional or social diversity is rare, so we also include other socio-spatial factors (Table 2). While putting them together, many mixed and inconsistent results can be found. For example, Powe et al. (2016) find that SD of building age has significantly positive relations with Simpson Index of race in Seattle, but no significant relation is found in San Francisco, Washington DC, and Tucson. Relations between SD of building age and percentage of small economy are also mixed among four cities.

Many results show the role of built-form diversity is minor or even inverse with expectation. For example, Martino et al. (2021) find that diversity variables are less important than intensity and centrality variables, and Simpson index of building age is inversely related to Shannon entropy of socioeconomic variables and affordability variables. Powe et al. (2016) find that SD of building age has a significantly negative association with Herfindahl–Hirschman Index of occupation.

Quantitative evidence examining built-form diversity and other socio-spatial factors are insufficient to get rigorous conclusions. Complex socio-spatial relation is the nature, so we need more quantitative evidence that systematically examines the effect of built-form diversity, especially in different regions and urban contexts. Previous studies find that fine grain, well connectivity, and mid to high density may host socioeconomic diversity. But how diverse grain sizes, connectivity, and densities are related to functional and socioeconomic diversity? The effects of built-form diversity are underexplored comparing to other built-form measures (e.g., density, connectivity) and other built-environment sectors (e.g., land use, housing).

Table 2. Quantitative evidence relating built-form diversity with other factors.

Element	Quantitative evidence
Building	• SD of building age has significantly negative association with median resident age, Simpson Index of resident age (Powe et al., 2016); no significant relation with walking activities (Sung et al., 2015) • Shannon entropy of building age has positive correlation with Shannon entropy of local economic type and education attainment (Carlucci, Zambon, & Salvati, 2020). • Simpson index of building age is negatively related to the sum of Shannon entropy of education attainment, income, and race, and affordability index (Martino et al., 2021) • Entropy of building floor area approaches significance with pedestrian count (Ewing & Clemente, 2013) • Building typologies have significant variation in pedestrian flow (Berghauser Pont, Stavroulaki, & Marcus, 2019).
Plot	• Simpson index of plot area positively relates to affordability index (Martino et al., 2021) • SD of land patch area is highly correlated with land-use diversity, and it is among highest loading variables of factors (Song, Gordon-Larsen, et al., 2013)
Street	• Shannon entropy of street orientation relates to higher node degree (ways per intersection), more 4-way intersections, fewer dead ends, and average circuity (Boeing, 2019)

4. Discussion and conclusions:

In this paper, we summarize multidisciplinary definitions of diversity and existing studies measuring built-form diversity. A comprehensive definition of diversity deals with both category and object, which leads to three concepts – Variety, Balance, and Disparity. Within quantitative studies about built-form diversity, few measure diversity considering the three concepts at the same time. For those who claim to measure diversity, SD, CV, Simpson Index, or Shannon Entropy are taken as proxies for diversity measurement. More other studies do not aim for diversity, thus do not intend to develop comprehensive diversity measurement. Building age and plot area are the most common elements and

features that are measured. There are some quantitative evaluations on associations between built-form diversity and other socio-spatial factors, but they are still limited, comparing to other built-form measures and other built-environment sectors. The effects of built-form diversity shown in current results are somewhat weak or inconsistent.

Several unsolved questions are worth to be explored in future research. Firstly, current evidence shows overall weak associations between built-form diversity and other socio-spatial factors. However, are their associations well examined, given that measurements on built-form diversity are not comprehensive? Secondly, such a situation calls for integration of measurements on Variety, Balance, and Disparity, and application on different built-form elements and features. Although existing measurements seems complementary, there are challenges in integrating them. That is to say, how distance measures, which yield typologies, could be better integrated with measures transforming entities into numbers? Thirdly, although most existing studies focus on the neighborhood scale, there are debates on which scale of diversity we should pursue (Stanley, 2012; Talen, 2006b). Cross-scale comparison on associations between different dimensions of diversity is meaningful. Additionally, measurement on built-form diversity and other dimensions of diversity (e.g., land use mix) can provide methodological reference for each other.

References

1. ASBEC. (2017). Design for People Diverse. *Creating Places for People: An Urban Design Protocol for Australian Cities*. Retrieved from https://urbandesign.org.au/protocol-framework/principles/diverse/
2. Berghauser Pont, M., Stavroulaki, G., Bobkova, E., Gil, J., Marcus, L., Olsson, J., . . . Legeby, A. (2019). The Spatial Distribution and Frequency of Street, Plot and Building Types Across Five European Cities. *Environment and Planning B: Urban Analytics and City Science*, 46(7), 1226–1242. doi:10.1177/2399808319857450
3. Berghauser Pont, M., Stavroulaki, G., & Marcus, L. (2019). Development of Urban Types Based on Network Centrality, Built Density and Their Impact on Pedestrian Movement. *Environment and Planning B: Urban Analytics and City Science*, 46(8), 1549–1564. doi:10.1177/2399808319852632
4. Bobkova, E., Berghauser Pont, M., & Marcus, L. (2019). Towards Analytical Typologies of Plot Systems: Quantitative Profile of Five European Cities. *Environment and Planning B: Urban Analytics and City Science*. doi:10.1177/2399808319880902
5. Bobkova, E., Marcus, L., & Berghauser Pont, M. (2017). *Multivariable Measures of Plot Systems: Describing the Potential Link between Urban Diversity and Spatial Form Based on the Spatial Capacity Concept*. Paper presented at the Proceedings of the 11th Space Syntax Symposium.
6. Boeing, G. (2019). Urban Spatial Order: Street Network Orientation, Configuration, and Entropy. *Applied Network Science*, 4(1), 67. doi:10.1007/s41109-019-0189-1
7. Bourdic, L., Salat, S., & Nowacki, C. (2012). Assessing Cities: A New System of Cross-Scale Spatial Indicators. *Building Research & Information*, 40(5), 592–605. doi:10.1080/09613218.2012.703488
8. Carlucci, M., Zambon, I., & Salvati, L. (2020). Diversification in Urban Functions as a Measure of Metropolitan Complexity. *Environment and Planning B: Urban Analytics and City Science*, 47(7), 1289–1305. doi:10.1177/2399808319828374
9. Delclòs-Alió, X., & Miralles-Guasch, C. (2018). Looking at Barcelona through Jane Jacobs's Eyes: Mapping the Basic Conditions for Urban Vitality in a Mediterranean Conurbation. *Land Use Policy*, 75, 505–517. doi:10.1016/j.landusepol.2018.04.026
10. Dibble, J., Prelorendjos, A., Romice, O., Zanella, M., Strano, E., Pagel, M., & Porta, S. (2017). On the Origin of Spaces: Morphometric Foundations of Urban Form Evolution. *Environment and Planning B: Urban Analytics and City Science*, 46(4), 707–730. doi:10.1177/2399808317725075
11. Ewing, R., & Clemente, O. (2013). *Measuring Urban Design: Metrics for Livable Places*. Washington, DC: Island Press.
12. Fainstein, S. S. (2005). Cities and Diversity: Should We Want It? Can We Plan for It? *Urban Affairs Review*, 41(1), 3–19. doi:10.1177/1078087405278968
13. Feliciotti, A. (2018). *Resilience and Urban Design: A Systems Approach to the Study of Resilience in Urban Form - Learning from the Case of Gorbals*. (PhD). University of Strathclyde, Glasgow.
14. Fleischmann, M., Romice, O., & Porta, S. (2020). Measuring Urban Form: Overcoming Terminological Inconsistencies for a Quantitative and Comprehensive Morphologic Analysis of Cities. *Environment and Planning B: Urban Analytics and City Science*. doi:10.1177/2399808320910444
15. Jacobs, J. (1961). *The Death and Life of Great American Cities*. New York: Random House.
16. Junge, K. (1991). The Geometric Representation of Similarity and Dissimilarity. Extension to Measures of Dispersion and Comments on the Distance Model of Subjective Dissimilarity. *Scandinavian journal of psychology*, 32(2), 154–163.
17. Junge, K. (1994). Diversity of Ideas about Diversity Measurement. Scandinavian Journal of Psychology, 35(1), 16–26.
18. Louf, R., & Barthelemy, M. (2014). A Typology of Street Patterns. *Journal of The Royal Society Interface*, 11(101), 20140924.
19. Magurran, A. E. (2005). Biological Diversity. *Current Biology*, 15(4), R116–R118. doi:10.1016/j.cub.2005.02.006
20. Maly, M. T. (2000). The Neighborhood Diversity Index: A Complementary Measure of Racial Residential Settlement. *Journal of Urban Affairs*, 22(1), 37-47. doi:10.1111/0735-2166.00038
21. Marcus, L. (2010). Spatial Capital. *The Journal of Space Syntax*, 1(1), 30–40.

22. Martino, N., Girling, C., & Lu, Y. (2021). Urban Form and Livability: Socioeconomic And Built Environment Indicators. *Buildings and Cities*, 2(1). doi:10.5334/bc.82
23. McGarigal, K., Cushman, S., Neel, M., & Ene, E. (2002). FRAGSTATS: Spatial Pattern Analysis Program for Categorical Maps (Version v3): University of Massachusetts, Amherst. Retrieved from http://www.umass.edu/landeco/research/fragstats/fragstats.html
24. Page, S. E. (2010). *Diversity and Complexity* (Vol. 2): Princeton University Press.
25. Powe, M., Mabry, J., Talen, E., & Mahmoudi, D. (2016). Jane Jacobs and the Value of Older, Smaller Buildings. *Journal of the American Planning Association*, 82(2), 167–180.
26. Preservation Green Lab. (2014). *Older, Smaller, Better: Measuring How the Character of Buildings and Blocks Influences Urban Vitality*. Retrieved from https://forum.savingplaces.org/connect/community-home/librarydocuments/viewdocument?DocumentKey=83ebde9b-8a23-458c-a70f-c66b46b6f714
27. Rosner, T., & Curtin, K. M. (2015). Quantifying Urban Diversity: Multiple Spatial Measures of Physical, Social, and Economic Characteristics. In *Computational Approaches for Urban Environments* (pp. 149–181): Springer.
28. Salat, S., Bourdic, L., & Nowacki, C. (2010). Assessing Urban Complexity. *International Journal of Sustainable Building Technology and Urban Development*, 1(2), 160–167. Retrieved from http://www.urbanmorphologyinstitute.org/resources/assessing-urban-complexity/
29. Salazar-Llano, L. (2019). *Portraying Urban Diversity Patterns Through Exploratory Data Analysis*. (PhD). Universitat Politècnica de Catalunya, Barcelona. Retrieved from https://www.researchgate.net/profile/Lorena-Salazar-Llano/publication/340266593_Portraying_urban_diversity_patterns_through_exploratory_data_analysis/links/5e80f208299bf1a91b89f2aa/Portraying-urban-diversity-patterns-through-exploratory-data-analysis.pdf
30. Song, Y., Gordon-Larsen, P., & Popkin, B. (2013). A National-Level Analysis of Neighborhood Form Metrics. *Landscape and Urban Planning*, 116, 73–85. doi:10.1016/j.landurbplan.2013.04.002
31. Song, Y., Merlin, L., & Rodriguez, D. (2013). Comparing Measures of Urban Land Use Mix. *Computers, Environment and Urban Systems*, 42, 1–13. doi:10.1016/j.compenvurbsys.2013.08.001
32. Stanley, B. W. (2012). An Historical Perspective on the Viability of Urban Diversity: Lessons from Socio-Spatial Identity Construction in Nineteenth-Century Algiers and Cape Town. *Journal of Urbanism: International Research on Placemaking and Urban Sustainability*, 5(1), 67–86. doi:10.1080/17549175.2012.661639
33. Stirling, A. (2007). A General Framework for Analysing Diversity in Science, Technology and Society. *Journal of the Royal Society, Interface*, 4(15), 707–719. doi:10.1098/rsif.2007.0213
34. Sung, H., Lee, S., & Cheon, S. (2015). Operationalizing Jane Jacobs's Urban Design Theory: Empirical Verification from the Great City of Seoul, Korea. *Journal of Planning Education and Research*, 35(2), 117–130. doi:10.1177/0739456X14568021
35. Talen, E. (2006a). Design for Diversity: Evaluating the Context of Socially Mixed Neighbourhoods. *Journal of Urban Design*, 11(1), 1–32. doi:10.1080/13574800500490588
36. Talen, E. (2006b). Design That Enables Diversity: The Complications of a Planning Ideal. *Journal of Planning Literature*, 20(3), 233-249. doi:10.1177/0885412205283104
37. Talen, E. (2006c). Neighborhood-Level Social Diversity: Insights from Chicago. *Journal of the American Planning Association*, 72(4), 431-446. doi:10.1080/01944360608976764
38. Talen, E. (2008). *Design for Diversity*. Burlington: Routledge.
39. Talen, E., & Lee, S. (2018). *Design for Social Diversity* (2 ed.). New York: Routledge.
40. Vialard, A. (2013). *A Typology of Block-Faces*. (PhD). Georgia Institute of Technology, Atlanta. Retrieved from https://smartech.gatech.edu/bitstream/handle/1853/52182/VIALARD-DISSERTATION-2013.pdf
41. Weitzman, M. L. (1992). On Diversity. *The Quarterly Journal of Economics*, 107(2), 363–405. doi:10.2307/2118476

INTERNATIONAL FORUM ON URBANISM

Urbanism is a way of thinking and acting needed for sustainable development; urbanists are becoming inter- and transdisciplinary, bringing science, technology, and design into their field. As the world is in a cascading of crises related to, e.g. health, energy, food, and water, as well as issues of, e.g. traffic, housing, and services, integrated approaches become increasingly relevant as most of these challenges are related to each other. Having a holistic urbanism viewpoint is the path to resilience and sustainability. Here, we are not referring to a longer list of notions but particularly encouraging dialogues between (possibly perceived) conflicting discourses. This book is the proceedings of the 14th IFoU conference, a stepping stone towards a new culture of working together. Such a cultural change is needed to face the complexity and uncertainty of current and future challenges.

TU Delft BK Bouwkunde Department of Urbanism

TU Delft OPEN